BSCE Esoft Quiz Part-1

Question #: 1

A 45-year-old obese factory worker comes to the physician because of a painful swelling in his groin. He explains that it suddenly appeared three weeks ago after he lifted a heavy box from the floor. During surgical repair, the surgeon identifies an artery located lateral to the herniated bowel. This artery anastomoses with which of the following arteries at the posterior surface of the anterior abdominal wall?

A. Deep circumflex iliac
✓B. Superior epigastric
C. Inferior epigastric
D. Gonadal
E. Superficial circumflex iliac

Rationale: Correct : The inferior epigastric a. branch of external iliac a. and is used as a landmark in direct and indirect inguinal hernia. It anastomoses with the superior epigastric artery.
Incorrect: Superficial circumflex iliac a.(is branches of femoral artery)and deep circumflex iliac a.(is branch of external iliac artery). Gonadal artery (is branch of abdominal aorta).

Question #: 2

A 39-year-old primigravida comes to physician because of 3-week history of involuntary leakage of urine, when her intra-abdominal pressure increases during coughing, laughing, or lifting. Diagnosis of stress urinary incontinence is made. Which muscle is most likely affected to cause this condition?

A. Coccygeus
B. Obturator internus
✓C. Pubococcygeus
D. Piriformis
E. Psoas major

Rationale: The pubococcygeus muscle is part of the levator ani muscle which forms the pelvic diaphragm. The pelvic diaphragm stabilizes the organs in the pelvis including the bladder and urethra. If this muscle becomes damaged, then there can be urinary or fecal incontinence.

Question #: 3

A 27-year-old man comes to the physician because of a 4-day history of persistent coughing and productive yellow sputum. He has a history of frequent wheezing and dysphagia from childhood. A CT scan shows the right subclavian artery forming a vascular ring with the aorta, around the trachea and esophagus. Which of the following is most likely the embryonic mechanism behind the underlying pathology?

A. Persistence of the distal part of right dorsal aorta
B. Persistence of the right dorsal aorta with involution of the distal part of left dorsal aorta
✓C. Abnormal involution of the right 4th aortic arch artery and proximal right dorsal aorta
D. Constriction of the aortic arch distal to the ductus arteriosus

Rationale: Correct answer. Abnormal involution of the right 4th aortic arch artery and proximal right dorsal aorta is the correct answer. A vascular ring occurs when the aorta or its branches form a complete ring around the trachea and the esophagus.
Incorrect answer. Persistence of the distal part of right dorsal aorta is incorrect. If there is a persistence of the right dorsal aorta it would result in double dorsal aorta.
Persistence of the right dorsal aorta with involution of the distal part of left dorsal aorta would result in presence of aorta on the right side of the vertebral column.

Constriction of the aortic arch distal to the ductus arteriosus is incorrect. Constriction of aorta distal to ductus arteriosus would result in Coarctation of the Aorta

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Question #: 4

A 39-year-old woman presents to her primary care physician with severe back pain that began after lifting her 2-year-old daughter. The physician suspects a herniated disc and orders an MRI which is shown. Which of the following nerves would most likely be compressed in this injury?

A. L2
B. L3
✓C. L4
D. L5
E. S1

Rationale: Correct: Disc herniation has taken place between L3 and L4 level. The nerve exiting at that level is L3, but the nerve getting compressed will be L4.
Incorrect: In lumbosacral region all the spinal nerves exit below the named vertebra, but the next spinal nerve gets compressed in disc herniation.

Attachment:

For question 162748.png

Question #: 5

A 63-year-old man comes to the physician because of a 6-month history of low back pain that has not been relieved by pain medication. Physical examination shows tenderness to palpation in the lumbosacral region. The results from the imaging studies is shown in the attached image. Which of the following is the most likely diagnosis?

A. Spina bifida cystica

B. Lumbar disc-herniation

C. Spondylolysis

D. Spondylolisthesis

✓E. Spondylolytic spondylolisthesis

Rationale: Correct: X-ray shows Traumatic spondylolysis (Fracture of the column bones connecting the superior and inferior articular processes) and spondylolisthesis(anterior displacement of vertebral body compared to inferior counterpart) of L4.
Incorrect: Lumbar disc herniation will be more visible on CT or MRI and spina bifida denotes non-fusion of the halves of the embryonic neural arches.

Attachment:

FOR QUESTION 162754.jpg

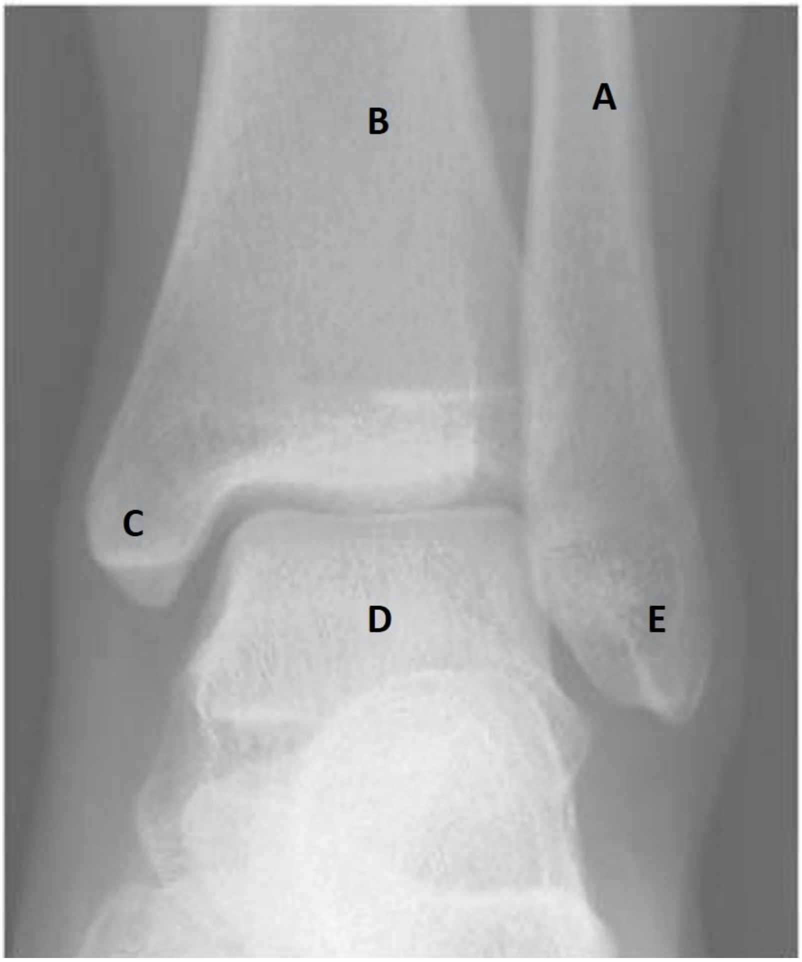
Question #: 6

A 23-year-old woman is brought to the emergency department because of a 4-hour history of severe pain after forceful eversion of her ankle. Physical examination shows a swollen, tender ankle joint with a limited range of motion. An x-ray of the ankle shows no fractures. MRI of the ankle joint shows the deltoid ligament has been torn from the medial malleolus. In the attached image, which of the following labeled areas most likely corresponds to the injured bony structure?

A. A
B. B
✓C. C
D. D
E. E

Rationale: The image shows the ankle joint. The medial, larger bone, is the tibia. The distal end of the tibia (C) forms the medial malleolus. This can be palpated on the medial aspect of the ankle joint.

Attachment:



Ankle.jpg

Question #: 7

A 10-year-old boy is brought to the emergency department by his parents because of a 2-hour history of an injury to his left foot. Physical examination shows the wound is on the medial aspect of the sole of his foot, just distal to the calcaneus. Damage to the medial plantar nerve is suspected. Which of the following actions would most likely be lost if there is damage to this nerve?

✓A. Abduction of the great toe
B. Flexion of the great toe
C. Adduction of the great toe
D. Flexion of all of the toes
E. Spreading and bringing together all of the toes

Rationale: The medial plantar nerve innervates the LAFF muscles of the feet. These are the abductor hallucis, flexor digitorum brevis, flexor hallucis brevis and the 1st lumbrical. The patient is still able to flex his toes including the great toe, because the flexor hallucis longus and flexor digitorum longus still have their innervation (tibial nerve in the leg).

Question #: 8

A 24-year-old man comes to the emergency department because of a 3-hour history of a stab wound just above and posterior to the medial malleolus. Physical examination shows loss of sensation on the entire sole of the foot and inability to spread the toes and bring them back together. He also has weakness of flexion of the toes. Which of the following nerves is most likely damaged?

A. Superficial fibular
B. Lateral plantar
C. Deep fibular
✓D. Tibial
E. Medial plantar

Rationale: The tibial nerve passe through the tarsal tunnel posterior to the medial malleolus. It then divides into the medial and lateral plantar nerves which innervates the intrinsic muscles of the feet and gives cutaneous sensation to the sole of the feet.

Question #: 9

A carpenter suffered a deep cut to the tip of his thumb. Initially the redness, swelling and pain were limited to the injured part of the thumb, but later the entire thumb and thenar eminence became inflamed. Which group of lymph nodes is the first to receive drainage from this injury?

A. Posterior axillary
B. Apical axillary
✓C. Lateral axillary
D. Anterior axillary
E. Central axillary

Rationale: he lateral (brachial) axillary lymph nodes are the primary nodes responsible for lymphatic drainage from the upper limb, including the thumb. Therefore, after a thumb injury, these nodes would be the first to receive drainage and respond to any resulting inflammation or infection.

Question #: 10

A 40-year-old man is brought to the emergency department after he was involved in a motor vehicle collision. Physical examination shows severely weakened flexion of the forearm and loss of sensation on the lateral surface of the forearm. Imaging studies show multiple fractures of the humerus. Which of the following additional findings will most likely be present in this patient?

A. Weakness in adduction of the wrist
✓B. Weakness in supination of the forearm
C. Difficulty in pronation of the forearm
D. Loss of sensation on the medial surface of the arm
E. Difficulty in extension of the forearm

Rationale: The muscles of the anterior compartment of the arm are responsible for flexing the forearm. These include the biceps brachii and brachialis muscles. Severely weakened flexion of the forearm and loss of sensation on the lateral surface of the forearm indicates that the musculocutaneous nerve is damaged. In addition to flexion of the forearm, the biceps brachii muscle also supinates the forearm. Therefore, if the musculocutaneous nerve is damaged, supination will also be affected.

Question #: 11

A 52-year-old woman comes to the physician because of a 6-month history of a lump in her left breast. Physical examination shows inversion of the left nipple, while the right is normal. A mammogram shows a 6-cm mass in the lower lateral quadrant. Which of the following structures is most likely to be affected, leading to the changes seen in this patient's nipple?

A. Branches of cutaneous pectoral nerves
B. Lactiferous ducts
C. Mammary arteries and veins
D. Mammary lymphatics
✓E. Suspensory (cooper) ligaments

Rationale: Cooper ligaments are the fibrous connections between the inner side of the breast skin and the pectoral muscles. Working in conjunction with the fatty tissues and the more fibrous lobular tissues, they are largely responsible for maintaining the shape and configuration of the breast. They play a major role in preventing breast ptosis.

Question #: 12

A 68-year-old man comes to the physician because of a 4-week history of numbness and paresthesia in the medial aspect of his left leg. Six weeks ago, he underwent a venous stripping procedure in which the great saphenous vein and its branches were removed. Which of the following nerves is most likely damaged?

✓A. Saphenous
B. Femoral
C. Superficial fibular
D. Sural
E. Deep fibular

Rationale: Saphenous nerve will be damaged, as it supplies the medial side of the lower limb.

Question #: 13

A 30-year-old man comes to the emergency department because of a 2-hour history of a painful and unstable right knee. Physical examination shows the findings in the photograph. Which of the following structures is most likely damaged?

A. Anterior cruciate ligament
B. Medial collateral ligament
✓C. Posterior cruciate ligament
D. Lateral collateral ligament
E. Patellar tendon

Rationale: As seen during physical examinations, patients may have knee instability or posterior sag sign. That will indicate posterior cruciate ligament damage
Attachment:

Picture3.png

Question #: 14

A 49-year-old woman comes to the physician because of a 6-month history of progressive bone pain and multiple fractures after minimal trauma. She has end-stage renal disease and has been receiving maintenance hemodialysis for the past 2 years. Physical examination shows tenderness over long bones. Laboratory studies are shown in the table. Which of the following best explains the cause of these findings?

A. Hypoparathyroidism
✓B. Decreased activity of 1-hydroxylase
C. Reduced formation of cholecalciferol from 7-dehydrocholesterol
D. Decreased exposure to sunlight
E. Decreased activity of 25-hydroxylase
F. Parathormone (PTH) secreting tumor

Rationale: The patient has renal failure, so renal 1-hydroxylase activity is low. Low levels of 1,25- DHCC would result in less absorption of calcium and phosphate from the diet. Serum calcium and phosphate levels will be low in patients with renal failure.
Hypoparathyroidism results in low serum calcium but increased serum phosphate levels. Hyperparathyroidism due to a PTH-secreting tumor would result in high serum calcium and low serum phosphate levels.
Attachment:

Parameter	Patient	Reference range
Serum calcium	8.0	8.4-10.2 mg/dL
Serum phosphate	2.5	3-4.5 mg/dL
Serum alkaline phosphatase (ALP)	350	20-70 U/ L

84436.jpg

Question #: 15

A 24-year-old man comes to the physician for a routine health examination. He feels well and has no significant medical history. He does not take any medications or supplements. Physical examination shows no abnormalities. Laboratory studies are shown in the table. Which of the following is the most likely cause of these laboratory findings?

A. Acute liver damage
B. Myocardial infarction
✓C. Enzyme induction due to alcohol
D. Bone disease
E. Acute pancreatitis

F. Muscular dystrophy

Rationale: The patient has an isolated increase in GGT, which indicates enzyme induction due to alcohol consumption. Amylase and lipase levels are elevated in acute pancreatitis. Alanine aminotransferase (ALT) levels are elevated in acute liver damage. In myocardial infarction, Troponin (I and T) and CK-MB levels are elevated. ALP levels are elevated in obstructive liver disease (gall bladder disorders) and bone disease. CK-MM levels are elevated in muscle diseases.

Attachment:

Alkaline phosphatase	Normal
CK-MM	Normal
CK-MB	Normal
γ-glutamyl transpeptidase (GGT)	Elevated
α-Amylase	Normal
Pancreatic lipase	Normal
Alanine aminotransferase	Normal

140465.jpg

Question #: 16

An investigator is studying the properties of a replicative DNA polymerase isolated from a newly discovered bacterial strain. In vitro, studies show that the enzyme is capable of synthesizing DNA in the 5′→3′ direction but lacks 3′→5′ exonuclease activity. Which of the following processes is most likely defective in this bacterial strain?

A. mRNA synthesis
✓B. Proof-reading during replication
C. Splicing and capping of mRNA
D. Translation and protein synthesis
E. Unwinding of double-stranded DNA

Rationale: The 3'-5' exonuclease activity of the enzyme allows for the incorrect base pair to be excised. This activity is known as proofreading. Loss of proofreading leads to DNA mutations. RNA polymerase is required for mRNA synthesis.
Splicing of RNA and capping are post-transcriptional modifications. U1-SNURP assists in mRNA splicing.
Translation and protein synthesis occur on the ribosomes. Peptidyl transferase activity on the ribosomes catalyzes the peptide bond formation during translation.

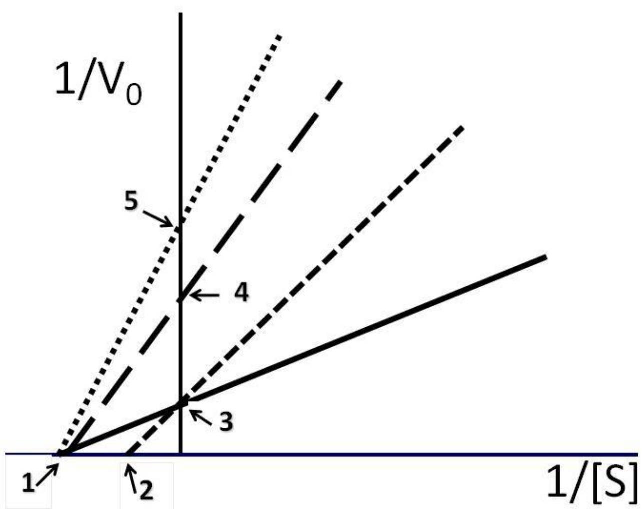
Helicase unwinds the double-stranded DNA during replication.

Question #: 17

An investigator is studying the enzyme kinetics of HMG-CoA reductase using a Lineweaver–Burk plot. The enzyme is analyzed under four conditions: without inhibitor (control), with statin, and with two other unknown inhibitors. Statin is a competitive inhibitor that is a structural analog of HMG-CoA and binds noncovalently to the enzyme's active site. The resulting Lineweaver–Burk plot is shown, with X- and Y-intercepts labeled 1–5. Which of the following points is best used to calculate the Vmax for the reaction in the presence of statin?

A. Point 1
B. Point 2
✓C. Point 3
D. Point 4
E. Point 5

Rationale: Statin is a reversible competitive inhibitor of HMG CoA reductase. The solid line (with the least slope) is the control without an inhibitor.
Point 1: Km of uninhibited reaction and Km in the presence of a non-competitive inhibitor
Point 2: Apparent Km in the presence of a competitive inhibitor
Point 3: (Correct Answer) Vmax of uninhibited reaction and Vmax in the presence of a competitive inhibitor. Note that in the presence of a competitive inhibitor, Vmax is the same, and apparent Km increases (lower affinity for substrate)
Point 4: Apparent Vmax in the presence of a low concentration of a non-competitive inhibitor. Note in the presence of a non-competitive inhibitor. Km is the same, and Vmax decreases.
Point 5: Apparent Vmax in the presence of a high concentration of a non-competitive inhibitor.
Attachment:



pic7.JPG

Question #: 18

A 20-year-old woman comes to the physician because of a 2-week history of gum bleeding while brushing her teeth. She has also noticed easy bruises on her arms and legs. She recently began a strict low-carbohydrate diet consisting mainly of meat, eggs, and cheese. She avoids fresh fruits and vegetables and does not take any vitamin supplements. Physical examination shows swollen, bleeding gums and small petechiae on the forearms. Which of the following functions is most likely defective in this patient?

A. Activity of lysyl oxidase
✓B. Activity of prolyl hydroxylase
C. Synthesis of fibrillin
D. Intestinal absorption of calcium and phosphate
E. Glycosylation of N-linked glycoproteins
F. Assembly of proteoglycans and glycosaminoglycans

Rationale: The vignette describes a patient with vitamin C deficiency. Vitamin C is required as a coenzyme for the post-translational modification of collagen, specifically hydroxylation of proline and lysine residues in collagen. Hydroxylation facilitates the formation of hydrogen bonds to form mature and stable collagen.
Copper is required as a coenzyme for lysyl oxidase, which is necessary to form stable covalent cross-links in collagen.
Calcitriol (1,25-DHCC) enhances dietary calcium and phosphorus absorption.

Question #: 19

A newborn male born at 28 weeks' gestation develops respiratory distress and cyanosis shortly after birth. He is intubated and admitted to the neonatal intensive care unit. A diagnosis of neonatal respiratory distress syndrome is made, and he receives exogenous surfactant replacement therapy containing artificial lung surfactant. Which of the following lipids is the most abundant component of the surfactant used in this patient?

A. Phosphatidylinositol containing arachidonic acid
B. Cholesterol esters containing stearic acid
✓C. Phosphatidylcholine with two saturated fatty acids
D. Sphingomyelin containing palmitate
E. Dipalmitoyl-phosphatidylethanolamine

Rationale: Lung surfactant contains dipalmitoyl phosphatidyl choline (DPPC). Also known as dipalmitoyl lecithin. It is a glycerophospholipid. It is produced in alveoli by type-II pneumocytes. Phosphatidyl inositol is a precursor of PIP2, which forms IP3 and DAG (second messengers).

Question #: 20

A 62-year-old man comes to the physician because of a 3-month history of needing frequent changes in his eyeglass prescription. He also reports increased thirst and urination. Laboratory studies show a fasting plasma glucose level of 180 mg/dL. A diagnosis of type 2 diabetes mellitus is made. Further studies indicate that his hyperglycemia results from decreased glucose uptake by skeletal muscle and adipose tissue due to insulin resistance. The glucose transporters responsible for uptake in these tissues normally mediate which of the following types of transport?

A. Primary active transport
B. Secondary active transport
C. Co-transport with Na^+ ions
✓D. Facilitated diffusion
E. Antiport with K^+ ions

Rationale: GLUT-4, present in the muscle and adipocytes, transports glucose along the concentration gradient and does not use energy. GLUT-4 is also insulin-responsive.
Na-K ATPase is a primary active transporter. Na-K ATPase is an antiporter.
SGLT is a secondary active transporter. SGLT transports glucose with Na (Co-transport). SGLT-1 is located primarily in the intestine (for dietary glucose absorption), and SGLT-2 is mainly located in the renal tubules for the reabsorption of filtered glucose.

Question #: 21

An investigator is studying the molecular mechanisms involved in polyglutamine diseases, such as Huntington disease. Analysis of affected cells shows aberrant transcriptional regulation associated with defective acetylation of histones. Which of the following best describes the relationship between histone acetylation and gene transcription?

A. Deacetylation of histones forms euchromatin
✓B. Deacetylation of histones reduces transcriptional activity
C. Acetylation increases the affinity between histones and DNA
D. Acetylation causes the recruitment of DNA methyltransferase
E. Acetylation activates and recruits topoisomerase

Rationale: Histones are positively charged proteins due to abundance of lysine, histidine, and arginine (Positively charged amino acids).
Acetylation of the histones results in unwinding DNA (less binding DNA to histones) and increased gene expression.
On the other hand, deacetylation of the histones results in increased binding of the negatively charged DNA, resulting in decreased expression of the involved genes.
Deacetylation facilitates the formation of heterochromatin and tight winding of the DNA around the histones.
This is a mechanism of regulation of gene expression in eukaryotic cells.

Question #: 22

A 2-year-old boy is brought to the physician by his mother because of repetitive self-injurious behavior, including biting of his fingers and lips. She reports that he has had profound developmental delays since early infancy. Physical examination shows global developmental delay, hypotonia, and self-mutilating behavior. Laboratory studies show:

- Markedly elevated serum uric acid

- Increased urinary uric acid excretion

Which of the following inherited enzyme deficiencies is the most likely cause of this patient’s condition?

A. Adenosine deaminase
B. Purine nucleoside phosphorylase
C. Orotate phosphoribosyl transferase
✓D. Hypoxanthine guanine phosphoribosyl transferase
E. Xanthine oxidase
F. PRPP synthetase

Rationale: The patient in the vignette has X-linked Lesch-Nyhan syndrome. Note the self-mutilation, hyperuricemia, and a delay in developmental milestones. HGPRT is an enzyme of the salvage pathway of purine nucleotides.
ADA deficiency results in low B- and T-lymphocyte function and severe combined immunodeficiency (SCID).
Orotate phosphoribosyl transferase (OPRT) deficiency results in megaloblastic anemia and developmental delay. Serum uric acid concentration is within normal limits.
Xanthine oxidase deficiency results in low serum and urinary uric acid concentrations.
Purine nucleoside phosphorylase (PNP) deficiency results in T-cell immunodeficiency.
PRPP synthetase and glutamine phosphoribosyl amido transferase are regulated enzymes of purine nucleotide synthesis. Increased activity of these enzymes results in hyperuricemia and increased risk of gout; however, developmental delay is not observed.

Question #: 23

An investigator isolates a mutant strain of Escherichia coli that demonstrates defective termination of transcription. Further analysis reveals a point mutation in a gene encoding a protein required for proper termination of transcription. Which of the following proteins is most likely to have reduced function in this bacterial strain?

A. DNA A protein
✓B. Rho factor
C. Single strand binding proteins (SSB)
D. Initiation factor-1 (IF-1)

E. Peptidyl transferase
F. Sigma factor

Rationale: Rho factor is the protein required for the termination of transcription. Mutations in the rho factor may cause defective termination of transcription. The sigma factor is necessary for the initiation of transcription. DNA protein and SSB are required for DNA replication, whereas IF-1 and peptidyl transferase are essential for translation (protein synthesis).

Question #: 24

A scientist studying transposable element (or mobile) DNA elements such as LINEs and SINEs makes an analogy that these mobile transposons are usually relatively harmless viruses. However, occasionally, a type of these transposons may insert into a critical gene causing disease. Of course, their activity is usually suppressed (by unknown mechanism), otherwise critical genes might be interrupted, and our genomes would be destroyed. What best explains how these DNA elements expanded in the human genome?

✓A. LINEs encode reverse transcriptase
B. LINEs propagate during meiosis
C. Transposons inhibit normal splicing events
D. Transposons hijack telomerase
E. LINEs are found within the telomere
F. LINEs are found within the centromere

Rationale: The Long Interspersed Nuclear Element (LINE) is a repetitive sequence in the human genome, as is the Short Interspersed Nuclear Element (SINE). The line contains the coding sequence for the cell to create reverse transcriptase. If RNA polymerase II creates an mRNA for a LINE, that mRNA will go to the cytoplasm where it may be translated into protein. If the reverse transcriptase enzyme is produced, the LINE mRNA may create a DNA copy, which may then go into the nucleus and reinsert somewhere else in the genome. The SINE is shorter (short), so it does not encode reverse transcriptase, but it is mRNA may also be reverse transcribed into DNA (if the RT is present), so the SINE may propagate in a similar manner to the LINE; it just needs a little help. It is important to understand some aspects of the human genome because more than 50 percent of our genome is repetitive sequence. Also, thankfully, the activity of these transposons is suppressed, especially during meiosis; however, it is estimated that in every generation approximately 17 transposition events do occur. If one of these transpositions interrupts an important gene, disease may result.

Question #: 25

A 3-year-old boy has hemolytic anemia. The hemolysis was due to accumulation of alpha-globin chains in erythrocytes. Which of these hemoglobin disorders is most likely in this patient?

A. Deletion of four alpha-globin genes

✓B. Mutations in both beta-globin genes
C. Mutations in both gamma-globin genes
D. A glutamate to valine substitution on sixth amino acid of beta-globin
E. A glutamate to lysine substitution on sixth amino acid of beta-globin

Rationale: The vignette describes a child with hemolytic anemia due to accumulation of alpha globin chains. Excessive accumulation of alpha-globin chains is observed in beta thalassemia. Due to mutations in both beta-globin genes, there is decreased production of beta-globin chains. As a result, there is excessive accumulation of alpha-globin chains which result in hemolysis.
Deletion of four alpha-globin genes results in hydrops fetalis and is not compatible with life.
Gamma-globin gene mutations are generally not severe as HbF is a minor Hb in adults.
Glu to Val substitution results in polymerization of Hb and sickling of RBCs.
Glu to Lys substitution results in HbC which is less soluble than HbA.

Question #: 26

A 10-year-old girl is brought to the physician by her parents because of a progressive accumulation of discolorations and growths on her skin. Physical examination suggests that the abnormal tissues are benign neurofibromas. Targeted genetic analysis of her *NF1* gene shows that she has a pathogenic mutation which she inherited through her germline. Testing of the parents shows maternal inheritance. The mother was diagnosed only after genetic testing because she was mildly affected. Which of the following genetic events is most likely to occur first during the manifestation of the numerous neoplasms in this patient?

A. Mutation of the mitochondrial genome
B. *NF1* increases apoptosis
C. Accumulation of toxic *NF1* RNA
D. *NF1* activates a cell cycle checkpoint
✓E. Loss of *NF1* heterozygosity

Rationale: NF1 is a TSG, when inherited in the germline, each cell in that person's body starts out with one "hit". When a mutation occurs in the other copy of this TSG, that cell now would have no functional copies of that TSG. Without any functional copies of a specific TSG, that cell would have a proliferative advantage, and be more likely to transit through the cell cycle, which contributes to cancer development and progression. The first step in the formation of the neoplasms in this patient would be loss of the other "good" copy of NF1. The cell that gets this "second hit", obtains a proliferative advantage against the other cells in the body, and is more likely to transit through the cell cycle. Now, it can accumulate mutations, possibly in other genes, such as protooncogenes, which would then become oncogenic.

Question #: 27

A 16-year-old boy is brought to the physician by his parents because he is not developing secondary sexual characteristics. Physical examination shows testicular atrophy, gynecomastia, and normal intelligence. Which of the following cytogenetic findings would be consistent with the most likely diagnosis of this patient?

A. A paracentric inversion on the X chromosome
B. An isochromosome X karyotype: 46,X,i(Xq)
✓C. The presence of one Barr body
D. An extra copy of chromosome 21
E. Maternal uniparental disomy of chromosome 15

Rationale: This man most likely has Klinefelter syndrome, and he would have a karyotype 47,XXY. One of his X chromosomes would be inactivated in an attempt to ensure balanced expression; this inactivated X would be visible as a Barr body. A paracentric inversion on the X chromosome would increase the likelihood that deletion or duplication events would occur in his offspring. A female with isochromosome X would have Turner syndrome. An extra copy of chromosome 21 (trisomy 21) is Down syndrome. Maternal uniparental disomy would cause Prader-Willi syndrome.

Question #: 28

A researcher stimulates a cell culture with insulin-like growth factor to investigate how gene expression is altered in response to this hormone. Poly-T conjugated sepharose beads are used to isolate mRNA by hybridization to the poly-A tail. Reverse transcriptase is used to synthesize single strands of DNA using the captured mRNA as template. Which of the following would best describe the DNA strands that are synthesized in this procedure?

A. They lack exon sequences
B. They contain intron and exon sequences
C. They are the promoter sequences
✓D. They lack intron sequences
E. They contain intron and promoter sequences

Rationale: Most human genes contain intervening sequences called introns, and these are normally spliced out post-transcriptionally, so a cDNA created from an mRNA by reverse transcriptase should lack intron sequences. The cDNA should contain exon sequences. The cDNA should not contain promoter sequences.

Question #: 29

A 52-year-old man is brought to a physician by his wife because of a 3-year history of progressively worsening chorea. His wife reports that he has had personality changes, slight loss of memory and gross sleep disturbances that began when he was in his mid-to-late forties. Which of the following best describes the genetic test that would be used to diagnose the disorder in this patient?

A. G-band karyotype
B. Direct enzyme assay
✓C. Sizing of PCR products
D. Northern analysis
E. Microarray CGH

Rationale: First you must recognize that the patient described in this vignette has symptoms suggestive of Huntington disease (HD) which is a trinucleotide repeat expansion disorder. In HD, a CAG codon found in exon 1 of the gene expands to a pathogenic number leading to a long repeat of glutamine amino acids in the encoded protein after translation. This expanded glutamine tract then causes the gene to attain some novel function which leads to neurodegeneration and the symptoms of HD. The trinucleotide repeat expansion can be directly measured by separation of PCR products (after amplification of this region). If the repeat is too long, it is considered pathogenic. G-band karyotype uses metaphase stage chromosome stained with Giemsa to visualize the chromosomes of a person. Enzyme assays look at the activity of enzymes and are very important in the diagnosis of many disorders (consider Tay Sachs disease). Northern blot analysis uses a DNA probe to detect the complimentary RNA sequence after transferring out of a gel and onto a nylon membrane. Microarray CGH detects copy number variation with a resolution of approximately 30 kb.

Question #: 30

A 39-year-old woman comes to the physician because of a 2-year-history of progressively worsening muscle weakness and a reduced ability to release her grip after squeezing. For instance, she has difficulty releasing a doorknob after opening the door. Genetic testing by PCR amplification of a repetitive sequence is diagnostic for a specific neurodegenerative disorder. Which of the following best describes the disorder caused by the pathogenic variant in this patient?

A. Myoclonic epilepsy
B. Muscular dystrophy
C. Multiple sclerosis
✓D. Myotonic dystrophy
E. Huntington disease
F. Fragile X syndrome
G. Neurofibromatosis

Rationale: This patient has myotonic dystrophy due to an expansion of a triplet repeat sequence in the DMPK gene. Myotonus means inability (or reduced ability) to relax after muscle contraction. A myoclonus is a sudden involuntary muscle twitch.

Question #: 31

An investigator is studying the telophase stage of mitosis where microtubules pull sister chromatids to opposite ends of the cell poles. Which of the following best describes the portion of the chromosome that attaches to the microtubules to drive this movement?

A. Euchromatin
✓B. Centromere
C. Telomere
D. The *Alu* sequence
E. Pseudogene

Rationale: Microtubules attach to the centromeric region of chromosomes, and the centromere is composed of repetitive sequences. The centromere does not contain expressed genes. Euchromatin describes areas of the genome that have active gene expression. Telomeres are repetitive sequences that are found on the termini of chromosomes. The Alu sequence describes a DNA restriction endonuclease site that tends to be found within some transposable elements such as the SINE (short interspersed nuclear element). There are millions of copies of this repetitive sequence in the genome. Pseudogenes are DNA sequences that look like genes, but for some reason are not expressed.

Question #: 32

A researcher studying non-syndromic congenital heart defects (CHD) finds that the population incidence of this multifactorial developmental defect is approximately 1 in 100 live births. Comparisons are made to estimate recurrence risk depending on the affected relative. Which of the following best explains an aspect of the inheritance of this disorder?

A. Recurrence risk will be higher in their male children then in their female children
✓B. Recurrence risk reduces dramatically as relatedness decreases
C. Recurrence risk will be the same as the population incidence
D. Recurrence risk will be one half of the population incidence
E. There is no risk to the children as this is a quantitative trait
F. Recurrence risk will be twice the population incidence

Rationale: A non-syndromic heart defect implies that there is not a “single-gene” or Mendelian genetic cause for the disorder. Instead, it is thought that there is likely to be a combination of many genetic factors and environmental influences to cause the disorder – multifactorial. An affected person who is more distantly related (say a cousin) contributes less risk than a first degree relative would. In other words, for multifactorial disorders, it matters more if a parent or sibling was affected, than a cousin. Some disorders such as pyloric stenosis exhibit a sex bias.

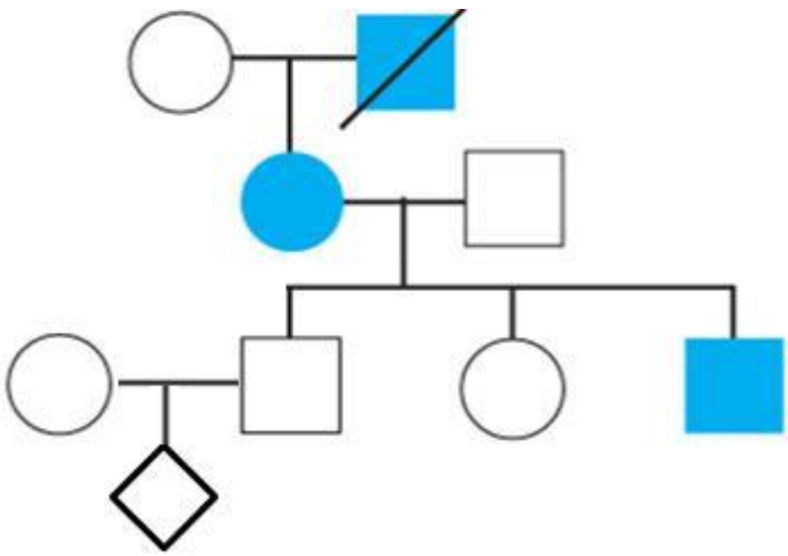
Question #: 33

A scientist is studying a rare genetic disorder to learn more about disease penetrance in people who inherit the disease-causing mutation. A representative pedigree of a family afflicted with the disorder is shown. Which of the following disorders is likely to be in this family?

A. Prader-Willi syndrome
B. Duchenne Muscular Dystrophy
✓C. Neurofibromatosis
D. Sickle Cell Anemia
E. Phenylketonuria

Rationale: NF is a classic autosomal dominant disorder with high penetrance, so we expect to see vertical inheritance of the trait. PWS is inherited only if the genetic aberration causes loss of expression from the paternal allele (either by maternal uniparental disomy, or mutation/deletion of the paternal copy). DMD is X-linked, and affected boys are usually unable to have their own children. SCD is autosomal recessive. Disorders caused by loss of function of an enzyme activity such as PKU, galactosemia, homocystinuria, are typically recessive. Note that since most genes are found on autosomes most inborn errors of metabolism (loss of enzyme activity) are autosomal recessive. Of course, some may be X-linked such as Lesch-Nyhan syndrome and G6PD deficiency.

Attachment:



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Question #: 34

A 32-year-old woman comes to the physician to learn about the recurrence risk for cystic fibrosis (CF) as she is planning to have a family. She has a niece who is a known carrier for a mutation causative of CF, and her husband is from a population where the incidence of CF is approximately 1 in 2,500. Which of the following is the best recurrence risk estimate of CF for the children of this couple?

A. 1/50
B. 1/40
✓C. 1/400
D. 1/2,500

E. 1/200

Rationale: The woman is at risk of having the same CF deficiency allele that her niece has. Since she has a niece who is a well-known carrier it means her sister has a 1/2 chance of being a carrier, and therefore her risk would be 1/4 to be a carrier (1/2 * 1/2). The husband is from the general population where carrier frequency is 1 in 2,500, so allele frequency is 1 in 50, and carrier risk would be 1 in 25. If they are both carriers they would have a 1 in 4 risk of having an affected child. Each probability is independent of each other, and they must all occur for the child to be affected, so we must multiply. So (1/4 * 1/4 * 1/25) = 1/400.

Question #: 35

A 63-year-old woman comes to the physician for a routine colonoscopy. During the procedure, a mass is observed in the ascending colon. A biopsy is taken, and a karyotype shows aneuploidy within the cells taken from the mass. The epithelium of the affected structure is usually a single cell layer thick with cells that are taller than they are wide. Which of the following best characterizes this epithelium?

✓A. Simple columnar
B. Transitional
C. Pseudostratified ciliated
D. Stratified cuboidal
E. Stratified squamous

Rationale: Aneuploidy is the abnormal number of chromosomes in a cell. Aneuploidy can originate during cell division when chromosomes do not separate properly & is regularly observed in malignant tumors.

The epithelium described here is that of simple columnar. Simple - single layer of cells. Columnar - cells are taller than they are wide.

Transitional: epithelium lining the lower urinary tract, it is a stratified epithelium, with specific morphologic characteristics that allow it to distend.
Pseudostratified ciliated: appears stratified, although some of the cells do not reach the free surface, all rest of the basement membrane.
Stratified cuboidal: more than one layer in the epithelium, but the most apical layer, the width, depth, and height of the cells are approximately the same.
Stratified squamous: more than one layer in the epithelium, but the most apical layer, the width of the cell is greater than its height.

Question #: 36

A 21-year-old woman comes to the physician because of a 6-month history of severe acne on her face. Physical examination shows some pimples are acutely inflamed and a few have some pus inside, which can be easily expressed out. The sebaceous glands involved in this condition have a secretory process wherein the entire cell becomes a part of the secretory product. Which of the following best describes this mode of secretion?

A. Merocrine
✓B. Holocrine
C. Autocrine
D. Paracrine
E. Apocrine

Rationale: Different types of exocrine glands can exhibit different mechanisms of secretion. In holocrine secretion, the secretory products accumulate in the maturing cells. The cells then undergo a type of programmed cell death releasing both the secretory products and cellular debris.
Merocrine secretion - vesicles from within the cytoplasm fuses with the apical plasma membrane and gets released into the extracellular space.
Apocrine secretion - the vesicles accumulate at the most apical part of the cell, and the apical part is eventually pinched off with those vesicles within it to release its contents to the extracellular space.
Autocrine and paracrine modes of secretion are seen in endocrine glands, not exocrine glands.

Sebaceous glands undergo holocrine mode of secretion.

Question #: 37

A 45-year-old man comes to the physician because of a 24-hour history of extensive skin lesions, fever, and dehydration. Physical examination of the skin lesions shows small vesicles which are easy to rupture. He is diagnosed with pemphigus vulgaris, which involves separation of cells within the stratum spinosum layer of the epidermis. Which of the following cell junctions are most likely affected in this condition?

A. Hemidesmosomes
B. Gap junctions
C. Zonula occludens
D. Zonula adherens
✓E. Desmosomes

Rationale: The cells of the stratum spinosum exhibit spinous cytoplasmic processes that attach to similar processes of adjacent cells by desmosomes.

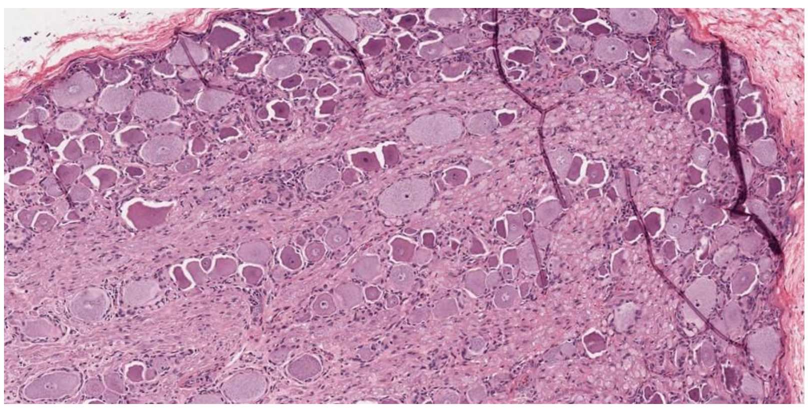
Question #: 38

A 17-year-old boy is brought to the emergency department because of a puncture wound to the mid back region after falling from a mango tree 30-minutes ago. Physical examination shows a penetrating wound adjacent to the vertebral column. He is admitted to the surgical ward for exploration. During surgery, a piece of tree branch is found lodged close to the spinal column and has damaged the structure indicated in the photomicrograph. The damaged structure is analogous to the nuclei of the CNS, in that it contains the cell bodies of neurons. Which of the following best describes the microscopic features of the damaged structure?

A. Synapsing occurring
B. Relatively small cell bodies
C. Cells with eccentric nucleus
✓D. Large cell bodies with thick bundles of nerve fibers
E. Cells with multiple dendrites and a single axon
F. Thin, randomly arranged nerve fibers

Rationale: The H&E slide shows neuronal soma arranged in neat lines interspaced by neuronal processes. This is the classic arrangement of dorsal root ganglia which is formed by the cell bodies of psuedounipolar somatic sensory neurons . Post-ganglionic sympathetic neurons form sympathetic ganglia which contain multipolar neurons. These will therefore have an irregular arrangement of cells and fibers. Purkinje cells are found in the cerebellum . Ventral motor neurons are found in the ventral horn and like the sympathetic and parasympathetic ganglia are multipolar neurons.

Attachment:



DRG.jpg

Question #: 39

A 45-year-old man comes to the physician because of a 3-month history of sharp pain in the right buttock and thigh. He is a mechanic. Physical examination shows positive straight leg raise test. MRI shows a herniation of the L4-L5 intervertebral disc. Which of the following is most likely a characteristic feature of the cartilage component of this structure?

A. Matrix contains only Type II collagen fibrils
✓B. Linear isogenous groups
C. Matrix contains abundant elastic fibers
D. Firmly attached perichondrium
E. High capability for repair
F. Undergoes both appositional and interstitial growth

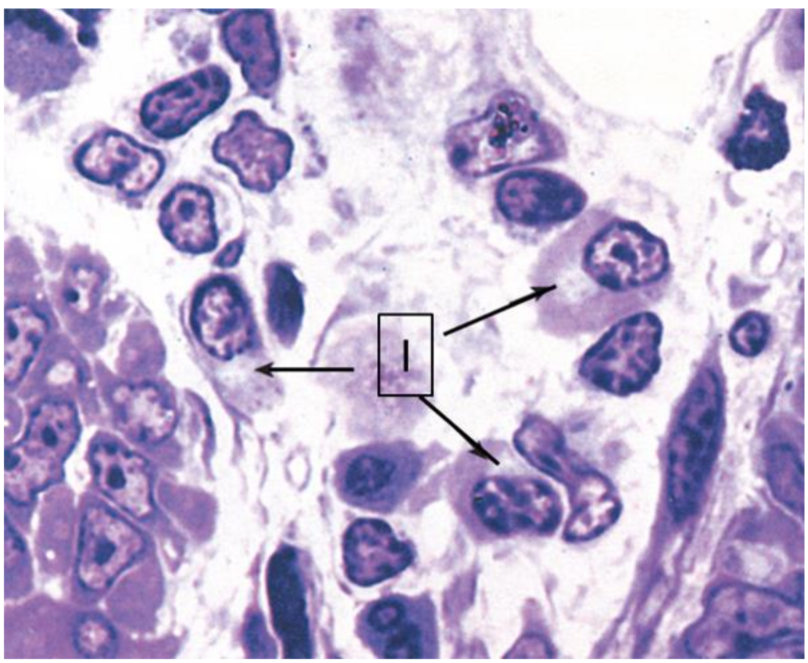
Rationale: Fibrocartilage consists of chondrocytes and their matrix in combination with dense connective tissue. It is typically present in intervertebral discs, the pubic symphysis, and articular discs.

Question #: 40

A 20-year-old man comes to the emergency department because of a 2-month history of fever and bone pains. Physical examination shows generalized lymphadenopathy and tenderness of the left tibia with an associated limp. An x-ray shows a mass within the tibia. Laboratory studies of blood show increased antibodies. Biopsy shows a tumor originating from the cells in the attached light micrograph. He is diagnosed with plasmacytoma. Which of the following organelles is most likely indicated by (I) in the light micrograph?

A. Rough endoplasmic reticulum
B. Smooth endoplasmic reticulum
✓C. Golgi complex
D. Lysosomes
E. Mitochondria

Rationale: Plasma cells secrete antibodies which require post transcriptional packaging by Golgi. Golgi complex is well developed in cells that secrete proteins by exocytosis. They appear as pale profiles in H&E and similar light microscopic images.
Attachment:



golgi .png

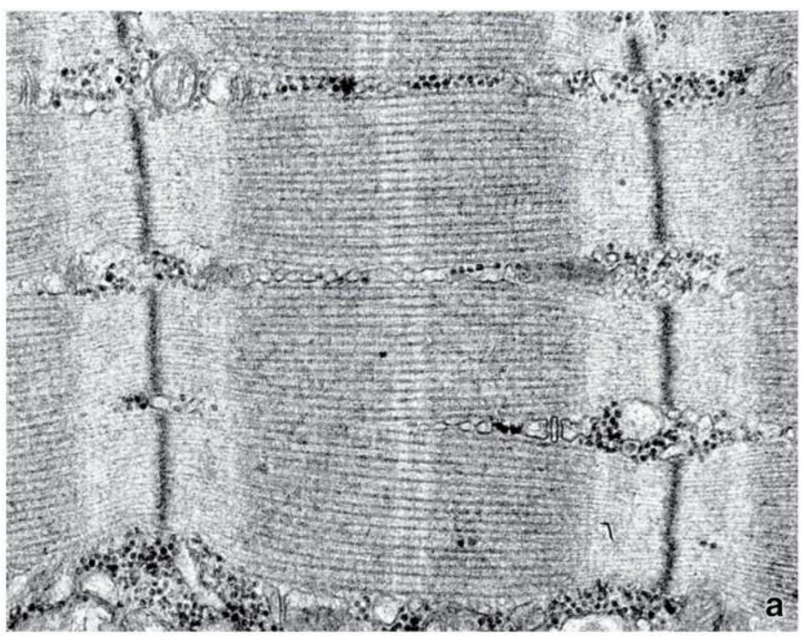
Question #: 41

A 17-year-old boy is brought to the physician by his parents because of a 4-month history of lower limb weakness. Physical examination shows atrophy of the muscles of both lower limbs and difficulty walking. A biopsy is obtained to rule out myopathy and is shown in the attached image. Molecular studies show antigen-antibody complexes against a protein that keeps thick filaments centered between the Z-lines of the sarcomere and prevents excessive stretching. Which of the following proteins is most likely affected?

A. Tropomyosin
✓B. Titin
C. Actin
D. Myosin
E. Desmin
F. Dystrophin

Rationale:
Titin is a large accessory protein that spans half of the sarcomere. Titin extends from the Z line and thin filament at its N terminus toward the thick filament and M line at its C terminus. Between the thick and thin filaments, two spring-like portions of titin help center the thick filament in the middle between two Z lines.

Attachment:



Sarcomere.jpg

Question #: 42

A 32-year-old man comes to the physician because of a 6-week history of progressively worsening weakened coordination of the arms (ataxia) and problems with speech articulation (dysarthria). He is currently undergoing chemotherapy and radiotherapy for small cell lung cancer. Brain CT scan shows atrophy of the cerebellum. Immunohistochemistry shows antibodies against Purkinje cells. He is diagnosed with paraneoplastic subacute cerebellar degeneration. Which of the following neurons best characterizes the targeted cell that is resulting in the symptoms seen in this patient?

A. Pseudounipolar
✓B. Multipolar
C. Golgi type II
D. Bipolar
E. Unipolar

Rationale: The cerebellar cortex has 3 layers, the outer or molecular layer - made up of stellate and basket cells, the inner (granular) layer - made up of granule and type II golgi cells and the middle or Purkinje layer - which is the target layer in this question. This is a single layer of motor neurons. Motor neurons are structurally multipolar.

Question #: 43

A 56-year-old woman is administered an intravenous dose of an antibiotic. Plasma drug concentrations are determined at 2 hours and 4 hours after administration. The drug concentrations are found to be 2 mg/L and 0.5 mg/L, respectively. The antibiotic follows first-order kinetics of elimination. Which of the following is most likely to be the half-life of elimination of the antibiotic?

A. 0.5 hours
✓B. 1 hour
C. 1.5 hours
D. 2 hours
E. 2.5 hours

Question #: 44

A 67-year-old man comes to the physician because of fatigue and lethargy. One month ago, he was diagnosed with stage 1 hypertension and started on propranolol. His pulse is 52/min, compared with 78/min one month ago. The physician switches him to pindolol. His symptoms resolve and his heart rate increases to 69/min. Which of the following properties of pindolol best explains its effect on this patient's heart rate?

A. It is a competitive antagonist
B. It is a full agonist
✓C. It is a partial agonist
D. It is an inverse agonist
E. It is a noncompetitive antagonist

Question #: 45

A 34-year-old man comes to the emergency department because of a 1-hour history of a drug overdose. The physician decides that the patient needs alkalinization of the urine. After the process is complete, there is a significant increase in the renal clearance of the drug. Which of the following pharmacokinetic properties most likely describes the causative drug?

A. The drug is a strong acid
✓B. The drug is a weak acid
C. The drug is a nonelectrolyte
D. The drug is a weak base
E. The drug is a strong base

Question #: 46

A 56-year-old man comes to the emergency department with a 5-year history of myocardial infarction, heart failure, and arrhythmia. He receives lidocaine by continuous IV infusion. The target plasma concentration is 3 mg/L. The pharmacokinetic parameters for lidocaine in the general population are Vd of 70 L, Clearance of 35 L/h, and half-life of 1.4 h. An infusion has begun. The plasma concentration of lidocaine is measured 2.8 h later and reported to be 2.4 mg/L. Which of the following is most likely to be the steady state plasma concentration in this patient?

A. 1.5 mg/L
B. 2.4 mg/L
✓C. 3.2 mg/L
D. 4.6 mg/L
E. 6.9 mg/L

Question #: 47

A 68-year-old woman comes to the physician because of a 2-week history of headaches. She is currently on prescription medication, but she cannot recall the name of the drug she is taking. Her blood pressure is 185/95 mm Hg. Physical examination shows tachycardia. Which of the following types of drugs is most likely to be her prescription medicine?

A. Alpha-1 adrenergic antagonist
B. Alpha-2 adrenergic antagonist
✓C. Beta-1 adrenergic agonist
D. Beta-2 adrenergic agonist

E. Muscarinic agonist

Rationale: Beta-1 receptors are located in the heart; therefore, stimulation of this receptor would lead to an increase in heart rate, heart force (contractility), and AV conduction velocity

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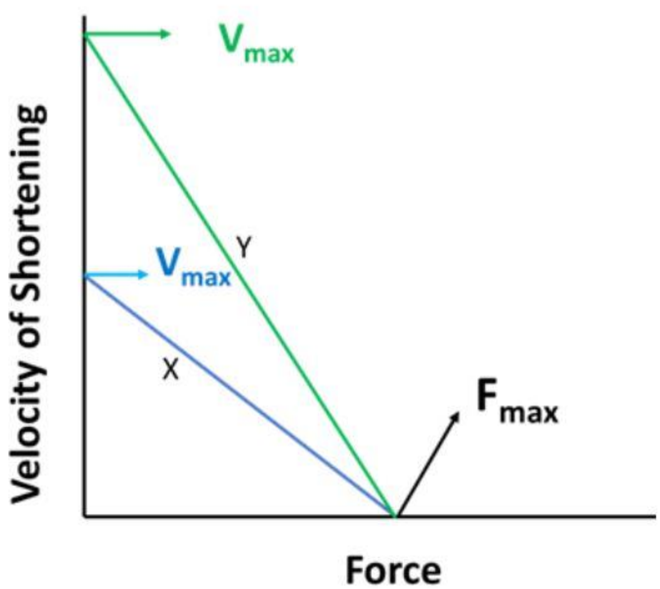
Question #: 48

A 30-year-old man comes to the physician for a routine examination. He is an athlete training for a competition, and over the past two months he has been training to improve the contractile force of his leg muscles by lifting a series of increased loads. The attached diagram represents the force velocity relationship from his leg muscles before (curve X) and after (curve Y) the 2 months of training. Which of the following explanations is the most likely reason for the shift from curve X to curve Y?

A. Decreased myosin ATPase activity
B. Increased afterload
✓C. Increased myosin ATPase activity
D. Increased preload
E. Recruitment of larger motor units

Rationale: C is the correct answer. Since the curve X shifts right to curve Y, Fmax is unaltered, but the Vmax is increased consequently. This shift happens when the myosin-ATPase activity increases in the muscle. Thus, curve X represents a slow, type I muscle fiber and curve Y represents a fast, type II muscle fiber. Thus, A is incorrect as decreased myosin ATPase activity will shift the curve X to the left, not to the right. B is incorrect as increased afterload will decrease the Vmax. B is incorrect as increased preload will increase Fmax. E is incorrect as recruitment of larger motor units would increase the Fmax.

Attachment:



Question #: 49

A 16-year-old boy is brought to the emergency room after being unable to walk and struggling to stand without help after competing at an athletics event. He says he feels extremely weak. He has a family history of primary hypokalemic periodic paralysis. Laboratory studies show that his plasma potassium level is 2.1 mmol/L (normal range 3.5 - 5 mmol/L). He is treated with a K^+ supplement in a saline IV infusion. Which of the following is the most likely primary cause of his weakness?

A. Depolarized rest membrane potential of excitable cells
✓B. Hyperpolarized resting membrane potential of excitable cells
C. Increased refractory period of voltage gated sodium channels
D. Depolarizing blockade of voltage gated sodium channels
E. Increased opening of delayed rectifier potassium channels

Rationale: At this level of severe hypokalemia, the Nernst equilibrium potential for potassium is significantly lowered (log(140/2.7) * -60 = -102 mV), which causes hyperpolarization of the membrane. This is the correct answer, depolarization would be the opposite. Refractory periods would decrease if there is any change. Voltage-gated sodium would be less likely to be blocked (in refractory state), and voltage-gated potassium channels would be less likely to be open.

Question #: 50

Nasal epithelial cells are isolated from human nasal brushings and bathed in an extracellular solution containing Na^+ 140 mM, K^+ 5 mM, Cl^- 105 mM. Assume normal intracellular electrolyte concentrations of Na^+ 14 mM, K^+ 140 mM, and Cl^- 40 mM. The membrane potential of one of the epithelial cells is measured and found to be -60 mV. Which of the following combinations best describes the driving force and direction of flux for sodium across the cell membrane?

A	120 mV	influx
B	120 mV	efflux
C	60 mV	influx
D	60 mV	efflux
E	30 mV	influx
F	30 mV	efflux

✓A. A
B. B
C. C
D. D

E. E
F. F

Rationale: Sodium gradient is 10-fold meaning the Nernst for Na is +60 mV. The actual Vm is -60 mV, therefore the driving force is the difference between Vm and Nernst for Na. This is 120 mV. Since the inside of the cell is negative, and the positive sodium ion has an inwardly directed concentration gradient as well as electrically attracted to the inside of the cell because of the cell negativity – this translates into an influx of Na+ ions into the cell.